

1	(a)	$S_y = u_y t + \frac{1}{2} a_y t^2$ $75 = 0 + \frac{1}{2} (9.81) t^2 \quad [\text{C1}]$ $t = 3.91 \text{ s} \quad [\text{A1}]$
	(b)	$v_y^2 = u_y^2 + 2a_y S_y$ $= 0 + 2(9.81)(75) \quad [\text{C1}]$ $v_y = 38.4 \text{ m s}^{-1} \quad [\text{A1}]$ or $v_y = u_y + a_y t$ $= 0 + (9.81)(3.91) \quad [\text{C1}]$ $v_y = 38.4 \text{ m s}^{-1} \quad [\text{A1}]$
	(c)	$S_x = u_x t$ $150 = u_x (3.91)$ $u_x = 38.4 \text{ m s}^{-1} \quad [\text{C1}]$ $v = \sqrt{38.4^2 + 38.4^2} \quad [\text{C1}]$ $= 54.3 \text{ m s}^{-1} \quad [\text{A1}]$ No, the package will not be damaged, since the impact velocity is less than 60 m s^{-1}. [A1]
	(d)	<u>Vertical acceleration is smaller</u> with air resistance. [M1] (Unchanged vertical displacement) Time of flight is larger. [A1]

2	(a)	Principle of moments states that for an object to be in rotational equilibrium, the
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